

# Cooper Le-Doux

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## Profile Summary

Biomedical engineering researcher specializing in EEG analysis, brain-computer interfaces, and data automation. Combined technical innovation with mentorship, outreach, and collaborative problem-solving. Proven ability to deliver measurable results in high-pressure lab, teaching, and interdisciplinary projects.

## Education

- **M.S. in Biomedical Engineering** (In progress, May 2027) – Univ. of Colorado Boulder
- **B.S. in Chemical Engineering** (Aug. 2020 – May 2025), GPA: 3.53 – New Mexico State Univ.
  - Minors: Biomedical Eng., Computational Eng., Electrical Eng., Chemistry

## Research Experience

### Lawrence Livermore National Laboratory – Research Intern

National Nuclear Security Administration Minority Serving Institutions Internship Program

- **EEG CNN Research (Aug 2023 – Aug 2025):** Developed an independent brain-computer interface project; improved signal processing accuracy and demonstrated self-directed problem-solving.
- **Cryptographic Analysis (Aug 2022 – Apr 2023):** Contributed to timing/thermal analysis; adapted quickly to new technical challenges.
- **Hydrogel Analysis (Jun 2022 – Aug 2022):** Automated microbead processing in MATLAB, reducing analysis time by 90% and increasing reproducibility of results.

### Neurable – EEG Analysis Project (Jun 2022 – Aug 2022)

- Analyzed and visualized EEG signals for brain-computer interface applications.

### Biophotonics & Flow Cytometry Lab, NMSU (Aug 2020 – May 2021)

- Hands-on experience in cell culture, microscopy, and flow cytometry techniques.

## Teaching & Outreach

### University of Colorado Boulder – Bio II Lab Teacher Assistant (Jan 2026 – May 2026)

- Guided undergraduate students through laboratory experiments in genetics, molecular biology, and physiology while ensuring proper lab technique and safety protocols.
- Assisted students in troubleshooting experiments, interpreting data, and understanding key

biological concepts during lab sessions.

- Graded lab reports and assignments, providing feedback to strengthen scientific writing, data analysis, and critical thinking skills.

### **Native Camp – Online English Instructor (May 2025 – Aug 2025)**

- Taught English to international learners, adapting to diverse ages, accents, and technical challenges.
- Strengthened communication, patience, and cultural awareness.

### **New Mexico Ranch & Heritage Museum – Blacksmith Volunteer (Aug 2024 – May 2025)**

- Engaged diverse audiences in demonstrations, ensuring safety and accessibility of technical knowledge.

### **Outdoor Recreation Center, NMSU (Jan 2020 – Jan 2022)**

- Assisted students with equipment rentals and bike repairs; developed problem-solving and customer service skills.

## **Food Service**

### **University of Colorado Boulder – Farrand Market Culinary Staff Member and Counter Attendant (August 2025 – December 2025)**

- Practiced safe food handling and cooking techniques
- Catered to customers' requests
- Managed high customer volume service
- Attended sales counter for shop merchandise
- Stocked new merchandise

## **Presentations & Publications**

Le-Doux, C. (Aug. 2022). Hydrogel Microbead MATLAB Code. Summer SLAM Poster Presentation, Lawrence Livermore National Laboratory.

## **Technical Skills**

- **Programming & Data:** Python, MATLAB, C, R, MNE-Python
- **Laboratory:** EEG Analysis, Microscopy, Cell Culture, Flow Cytometry
- **Other:** Technical Communication, Data Visualization

## **Languages and Certifications**

**Languages:** English (native), Spanish (secondary)

**Certifications:** Teaching English as a Foreign Language (TEFL)